Artificial Intelligence and RAG Guide
Document Overview: RAG (Retrieval-Augmented Generation) Architecture

1. Concept of RAG:
Retrieval-Augmented Generation combines LLMs with external knowledge bases.
Instead of relying solely on parametric training weights, RAG retrieves
relevant context at runtime and supplies it to the model prompt.

2. Key Benefits of RAG:
- Hallucination Reduction: Grounding responses in verified documents.
- Fresh Information: Keeping knowledge updated without expensive retraining.
- Verifiable Sources: Enabling explicit citations and page references.
- Security & Access Control: Keeping sensitive data in private databases.

3. Document Chunking Strategies:
Chunk size determines the size of text segments sent to the embedding model.
A typical chunk size ranges between 400 and 1000 characters.
Chunk overlap (usually 10 to 20 percent of chunk size) ensures that important
information at boundaries is not lost between adjacent chunks.

4. Vector Database and Cosine Similarity:
Embeddings map text to high-dimensional vectors (e.g., 3072 dimensions in Gemini).
Cosine similarity measures the angular distance between query and document vectors.
A similarity score close to 1.0 represents high contextual relevance.